



Adding and Subtracting Rational Expressions

As with numbers, rational expressions can be added and subtracted.

Adding or Subtracting Rational Expressions with Like Denominators

To add or subtract rational expressions with like denominators, add or subtract the numerators. Keep the same common denominator.

EXAMPLE 1

Add.

$$\frac{x^2 + 2x}{x - 3} + \frac{x^2 - 3x - 15}{x - 3}$$

SOLUTION

$$\begin{aligned}\frac{x^2 + 2x}{x - 3} + \frac{x^2 - 3x - 15}{x - 3} &= \frac{(x^2 + 2x) + (x^2 - 3x - 15)}{x - 3} \\ &= \frac{2x^2 - x - 15}{x - 3} \\ &= \frac{(2x + 5)(\cancel{x - 3})}{(\cancel{x - 3})} \\ &= 2x + 5\end{aligned}$$

Add the numerators.

Combine like terms.

Factor and simplify if possible.

Simplify. ■

Adding or Subtracting Rational Expressions with Unlike Denominators

To add or subtract rational expressions with unlike denominators, first rewrite the expressions with a common denominator, preferably the least common denominator. Then add or subtract. Just like with any fractions, the least common denominator of two rational expressions is the least common multiple of the denominators.

► **Think About It** Use these steps to find the least common denominator:

Step 1 Fully factor each denominator.

Step 2 Write each factor the greatest number of times it appears in any one denominator.

Step 3 Multiply the factors described in Step 2; the LCD is the product of those factors.

EXAMPLE 2

Subtract.

A $\frac{7x + 3}{(x + 1)(x - 3)} - \frac{5}{x + 1}$

SOLUTION

$$\frac{7x + 3}{(x + 1)(x - 3)} - \frac{5}{x + 1}$$

The denominators are in factored form.

The LCD is $\underbrace{x + 1} \cdot \underbrace{x - 3}$.
($x + 1$) appears once in the first and the second denominators. ($x - 3$) appears once in the first denominator.

$$= \frac{7x+3}{(x+1)(x-3)} - \frac{5}{x+1} \cdot \frac{x-3}{x-3}$$

The first ratio has the LCD. Multiply the second ratio by 1. For the appropriate form of 1, decide what factor(s) would be needed to get the LCD.

$$= \frac{7x+3}{(x+1)(x-3)} - \frac{5x-15}{(x+1)(x-3)}$$

Expand the second numerator.

$$= \frac{(7x+3) - (5x-15)}{(x+1)(x-3)}$$

Subtract the numerators.

$$= \frac{2x+18}{(x+1)(x-3)} = \frac{2(x+9)}{(x+1)(x-3)}$$

Simplify the numerator.

B
$$\frac{8}{5a^3 + 25a^2} - \frac{a+2}{a^3 + 10a^2 + 25a}$$

SOLUTION

$$\frac{8}{5a^3 + 25a^2} - \frac{a+2}{a^3 + 10a^2 + 25a}$$

$$= \frac{8}{5a^2(a+5)} - \frac{a+2}{a(a+5)(a+5)}$$

Factor the denominators so you can find the LCD.

The LCD is $\underbrace{5} \cdot \underbrace{a^2} \cdot \underbrace{(a+5)(a+5)}$.
 5 appears once in the first denominator.
 a appears twice (second power) in the first denominator and once in the second.
 (a + 5) appears twice in the second denominator.

$$= \frac{8}{5a^2(a+5)} \cdot \frac{a+5}{a+5} - \frac{a+2}{a(a+5)(a+5)} \cdot \frac{5a}{5a}$$

Multiply each ratio by 1. To determine the appropriate form of 1 for each ratio, decide what factor(s) would be needed to get the LCD as the denominator.

$$= \frac{8a+40}{5a^2(a+5)(a+5)} - \frac{5a^2+10a}{5a^2(a+5)(a+5)}$$

Expand the numerators.

$$= \frac{(8a+40) - (5a^2+10a)}{5a^2(a+5)(a+5)}$$

Subtract the numerators.

$$= \frac{-5a^2 - 2a + 40}{5a^2(a+5)(a+5)}$$

Simplify the numerator to get $-5a^2 - 2a + 40$, which cannot be factored. ■